

Folding Reality: Learning Spatial Computing through VR Interactive Paper Folding Game

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ABSTRACT

We introduce “Folding Reality” for conducting research on training spatial computing in virtual reality scenarios. It is designed for children and Large Multimodal Models (LMMs). We aim to understand how creativity emerges during the 2D to 3D folding process through interactive data gathered from children’s engagement with the game. We will then use this data to train LMMs, providing them an opportunity to enhance their spatial computing abilities through interactive paper folding. Our goal is to create a learning environment where both children and LMM can explore and understand spatial relationships, advancing research in this field.

Index Terms: Large Multimodal Model (LMM), Virtual Reality (VR), Creativity, Interactive Learning, Game-Based Learning.

1 INTRODUCTION

Virtual reality (VR) technology has established virtual environments as an effective learning platform [1]. These immersive experiences promote active engagement with content, allowing learners to acquire knowledge through practical application rather than passive observation [3]. “Folding Reality” is a VR game designed specifically for children and Large Multimodal Model (LMM). The game aims to advance research in spatial computing through interactive paper folding techniques. Unlike traditional methods that rely on three-view diagrams or static three-dimensional (3D) models, our approach helps learners actively transform two-dimensional (2D) objects into 3D structures.

We aim to understand how creativity emerges during the 2D to 3D folding process. This game creates a learning environment that enables children and LMM to develop spatial relationships independently, enhancing their creativity, spatial reasoning, and manual dexterity. This paper presents the design and implementation of our approach, demonstrating its potential to advance both educational practices and spatial computing research.

2 RELATED WORK

The Witness [2] is a puzzle game that challenges players to solve intricate puzzles by interacting with 2D grids and exploring 3D environments, promoting critical thinking and creativity through self-discovery and trial and error. Foldit, engages players in protein folding to address complex scientific challenges, allowing their solutions to assist researchers in identifying protein structures, thereby contributing to medical advancements [4]. Additionally, research by a Harvard team on reconfigurable metamaterials introduces a framework for designing materials that can autonomously switch functions based on their structure, simplifying the creation

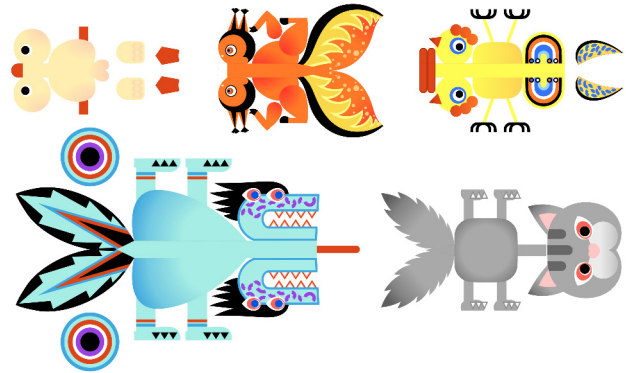


Figure 1: The graphic visual design and the other VR environments.

of versatile systems [5]. Inspired by Foldit’s success in integrating gaming with scientific inquiry and exploration games, we developed “Folding Reality” that promotes experiential learning and enhances spatial understanding through interactive manipulation of paper folding in a VR environment.

3 WORK OVERVIEW

“Folding Reality” transforms spatial learning through virtual paper folding, guiding users from 2D templates to 3D structures while serving two goals: developing children’s spatial reasoning and collecting interaction data to enhance LMMs’ spatial computing capabilities. We will collect data from children through workshops and observations through HTC Vive Focus 3 Eye Tracker, measuring variables such as folding sequences, completion times, hand movements, gaze patterns, and error rates to create a comprehensive dataset for educational research and LMM training.

Visual Design. The game’s visual design adapts Rorschach test principles of symmetric interpretation, featuring character templates with distinctive elements like eyes, body, and geometric patterns divided along a central axis (see Figure 1) to facilitate spatial learning [6]. This balanced approach ensures consistent folding logic while enabling systematic observation of player interaction patterns with standardized design elements.

VR Implementation. The VR application was developed using Unity 2021.3 LTS for its stability and excellent support for VR systems. We integrated the Unity XR Toolkit to manage interactions, including hand tracking for the paper-folding mechanics. Rather than relying on pre-designed animations, we implemented the core mechanics through real-time skeletal binding and model manipulation. This approach allows for dynamic and precise interactions, enhancing realism and giving players greater freedom in folding actions in the HTC Vive Focus 3.

Due to the limitations of interacting with virtual objects in a VR environment, such as difficulty in controlling the bending direction accurately, we implemented a predefined framework. The

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virtual paper model was equipped with a custom skeletal structure and then imported into Unity, where the control points of the skeleton were visualized (see Figure 2). This skeletal system maintains the paper's structural integrity while adapting to player manipulations. To simulate the behavior of real paper, user actions need to follow predefined crease lines. Folding actions are executed based on these crease lines, with the mesh adjusting instantly in real time in response to player interactions.

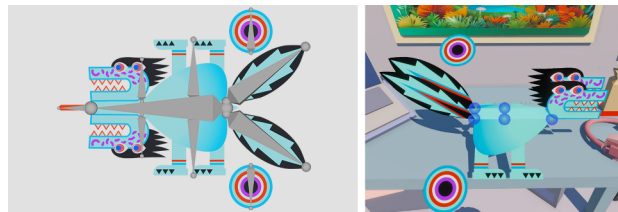


Figure 2: The model skeleton and control points.

We also detect gestures such as pinching and pulling using Unity's OpenXR framework. These gestures are mapped to specific skeletal movements, allowing players to manipulate the paper precisely. The implementation supports operations like creating folds by pinching and dragging along a crease line, flipping or rotating the paper, and undoing or refining actions. This seamless interaction between gesture recognition and skeletal control ensures an intuitive and immersive experience, closely replicating real-world paper-folding dynamics.

Large Multimodal Model as Player. LMMs have demonstrated impressive performance in solving complex reasoning questions [7], but they often struggle with procedural tasks that require spatial reasoning, such as paper folding. To enhance the spatial understanding of LMMs, we propose collecting interactive folding data within our VR game (from child-centered user study) and using it to train LMMs. Specifically, we decompose the gathered folding data into step-by-step action sequences. For each step, the input consists of the current state of the folded paper (with annotated control points) and the target 3D object. The output includes a manually annotated thought process and the correct action, specifying which action to execute, at which control points, and the rationale behind the action. By creating these training data pairs and applying LoRA fine-tuning, we allow the model to mimic children's learning-by-doing experience. This approach enables the model to conduct reasoning before performing valid sequential actions, transforming a flat 2D sheet into the final 3D object step-by-step (see Figure 3).

4 CHILD-CENTERED USER STUDY

Although our work is in its early stages, we have achieved the logic for virtual paper folding and are currently applying for IRB approval to conduct upcoming user studies with children.

This study will evaluate how interactive paper-folding tasks in a VR environment can enhance children's creativity, spatial reasoning, and manual dexterity. We focus on children aged 8 to 12, as this stage is crucial for spatial cognition and creativity. During this period, they move from concrete to formal operational thinking, enabling abstract reasoning while still benefiting from structured guidance. The study aims to understand their approach to the 2D-to-3D transformation process, measure improvements in spatial awareness, and identify factors that enhance engagement and learning effectiveness. A sample of 30 to 50 participants will be recruited from local schools and community programs to ensure diversity and reliability, with parental consent and child assent obtained to comply with ethical standards (Ethical protocols, including IRB approval and participant privacy safeguards, will be con-

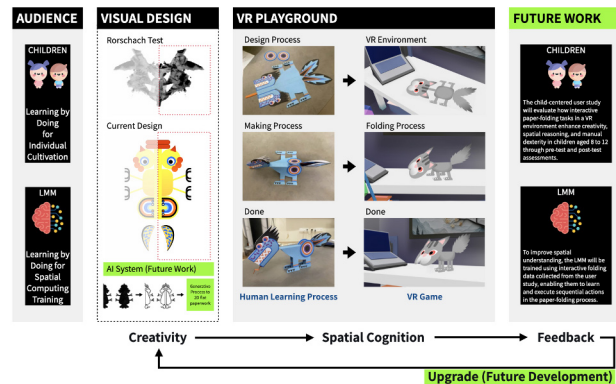


Figure 3: The workflow of "folding reality."

sidered for compliance and transparency as required by our university).

The study will employ pre-test and post-test frameworks to assess progress. Participants will complete a spatial reasoning test and a simple paper-folding task to establish baselines. Following five 30-minute VR sessions over two weeks, they will repeat similar tasks and provide feedback on their experience through questionnaires. Gameplay sessions consist of a tutorial phase for guided learning, a challenge phase with progressively complex tasks, and an exploration phase for creative experimentation. Data will be collected through in-game analytics, behavioral observations, and feedback from parents and teachers to evaluate performance, interaction patterns, and engagement.

5 CONCLUSION AND FUTURE WORK

"Folding Reality" has the potential to enhance spatial computing skills through interactive paper folding for children and LMM training. Using a local Stable Diffusion AI system in the future integrated into the VR environment, we can automate the generation of papercraft templates, transforming 2D silhouettes into detailed designs, while progressively expanding folding complexity from basic to advanced patterns. In addition, exposing this novel project will help us to gain community feedback before our formal study.

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